Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/20/2006
PI Name:	Bugbee, Bruce Ph.D.		
Project Title:	Ethylene synthesis and Sensitivity in Crop Plants		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	84322-4820	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-05
Start Date:	02/20/2002	**End Date:**	02/19/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Sensitivity. Understanding the threshold concentration below which ethylene has no effect is imperative to eliminating ethylene effects in the microgravity environment of spaceflight. Unfortunately, this threshold has never been reliably determined. Abeles (1992) suggested threshold values of 100 ppb for periodic exposure and 50 ppb for chronic exposure and these values have been cited by other authors (e.g. Stutte, 1992). Unfortunately, these threshold concentrations are rough estimates based on incomplete studies. Our studies indicate that lettuce, wheat, and rice are sensitive to ethylene levels of 25 ppb. Our most recent studies indicate that ethylene sensitivity is reduced by growing plants at higher temperatures. Synthesis. Healthy plants synthesize ethylene to mediate developmental stages from germination to senescence. However, elevated atmospheric ethylene levels cause a variety of abnormal responses including inhibited root and hypocotyl elongation, leaf epinasty, reduced growth, premature leaf senescence, and sterility (Abeles et al., 1992; Klassen and Bugbee, 2001; Mattoo and Suttle, 1991; Morison and Gifford, 1984; Smalle and Van Der Straeten, 1997).		

	Despite the extensive literature on biological ethylene production, rates of whole plant synthesis are not well characterized. Reported rates of synthesis range 20-fold from 0.1 to 2.0 nmol kg DW-1 s-1 in roots and shoots of healthy plants. Production rates are up to 10 times higher in stressed plants. Precise characterization of these synthesis rates is essential to the design of growth chambers for spaceflight. We are now characterizing ethylene synthesis rates in salad crops.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The need to identify the physiological processes limiting yield becomes more important as we minimize the environmental and cultural factors limiting yield (weeds, insects, diseases, water, and fertility). An understanding of whole plant physiology can be used to direct field research and can alter our preconceived ideas of what should be studied in the field. Whole plant studies provide a link between studies at the cellular, and single-leaf levels and communities of plants in the field. Integrating this knowledge into models that predict yield is a complex process. Higher Protein. One of our unexpected findings, for example, is that the protein concentration in our wheat is consistently higher than in the field, due perhaps to a more constant nitrogen concentration in the root zone or other interactions among environmental factors. Ultradwarf Cultivars. Semidwarf wheat is 80 to 100 cm tall. Full dwarf wheat is 50 to 70 cm tall. No commercial cultivars have been developed that are less than 50 cm tall because they do not compete with weeds and require high planting densities. However; dwarf wheats have less unproductive stem material and are thus capable of a higher harvest index than tall wheats. Higher harvest index usually results in higher yields. Dwarf wheats are also highly resistant to lodging, which can be a major problem. We do not have the conventional constraints to dwarf wheat production, and we have thus developed a series of ultradwarf cultivars for use in controlled environments. It may now be possible to demonstrate the yield advantages that have kept dwarf wheats from being used in the field. The challenge is to bring terrestrial wheat yields closer to what we have achieved in controlled environments.
Task Progress:	No progress report provided for this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)